

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT2 — VISUALIZATION DEVELOPMENT EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Development (BL4)	Assessment:	Assessment Tool 2 – Visualization Development
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO2 Statement:	<i>Analyze and synthesize multiple data views into a coherent, annotated dashboard that communicates a clear narrative insight. (BL4)</i>		
Student Name:	PRIYANNKA.G	Reg. No.:	192524124
Section / Batch:		Date:	30.07.2026

Code of Conduct

I Priyanka. G/ 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must include at least two coordinated charts sharing a common axis or theme.
- Include at least one annotation calling out a key data point in your dashboard.
- Write a one-sentence narrative takeaway summarizing the insight your dashboard reveals.

Annexure A – VISUALIZATION DEVELOPMENT

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 1 — Retail Quarterly Dashboard (5 Marks)

Given Data:

Quarter	Revenue (\$k)	Profit Margin (%)
Q1	200	15
Q2	250	18
Q3	220	12
Q4	300	20

Task: Build a two-chart dashboard: a bar chart of revenue and a line chart of profit margin, aligned on the same quarter axis. Add one annotation highlighting the quarter with the highest margin, and write a one-sentence takeaway summarizing the overall trend.

Dashboard Design (Key Elements):

Chart Types	Bar chart (Revenue) + Line chart (Profit Margin)
Shared Axis	Quarter (Q1 → Q4) — identical categorical/ordinal axis on both panels
Aesthetic Mapping	Bar colour = Quarter; line-point colour intensity = Revenue, linking the two panels
Annotation	Callout on Q4 identifying it as the quarter with the highest profit margin (20%)
Narrative Takeaway	One-sentence summary of the year's overall revenue and margin trend

Design Explanation / Working:

The dashboard places a bar chart of quarterly revenue directly beside a line chart of quarterly profit margin, with both panels sharing an identical Quarter (Q1–Q4) axis so that a viewer can visually align any quarter across the two charts without re-reading axis labels.

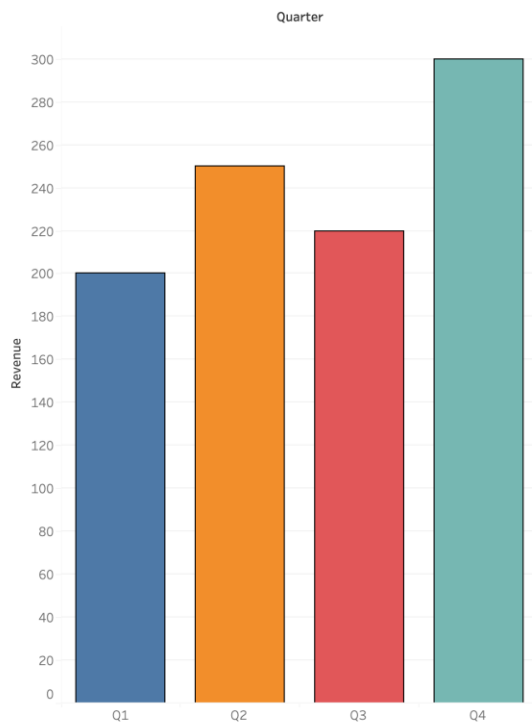
A secondary aesthetic a light-to-dark blue colour gradient applied to the line's markers is mapped to the Revenue value of that quarter, so the line chart is not only showing margin but also visually echoing the bar chart's revenue story: the darkest point (Q4) corresponds to both the tallest bar and the highest margin.

- Q1: Revenue = \$200k, Margin = 15% — baseline quarter.
- Q2: Revenue = \$250k, Margin = 18% — both metrics rise.
- Q3: Revenue = \$220k, Margin = 12% — revenue dips slightly and margin drops sharply, signalling a less profitable quarter despite reasonable sales.
- Q4: Revenue = \$300k, Margin = 20% — both metrics peak simultaneously.

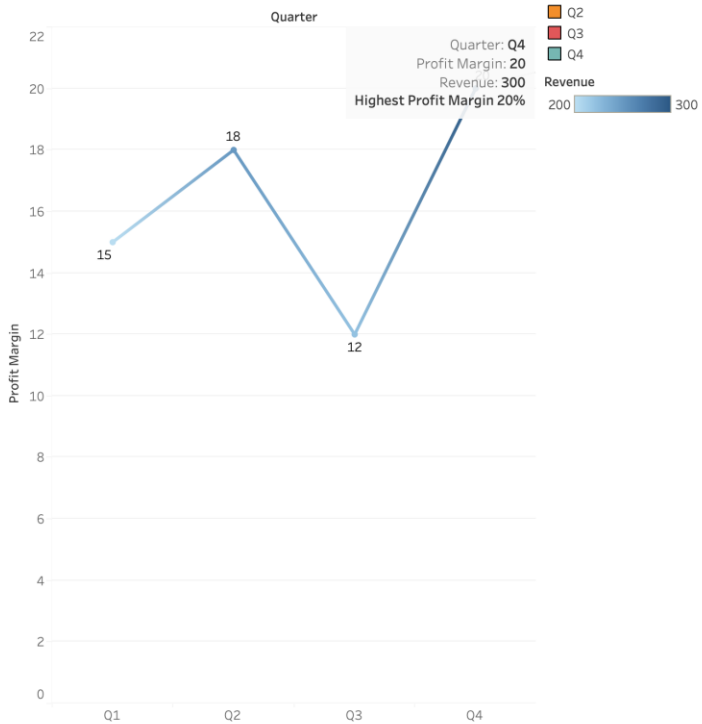
Constructed Dashboard:

192524124 - Retail Quarterly Dashboard

192524124 Quaterly Revenue Bar Chart



192524124 Quaterly Profit Margin Line Chart



Revenue and profit margin generally increased across the year, with Q4 achieving both the highest revenue and the highest profit margin.

Figure 1: Retail Quarterly Dashboard — Revenue (Bar) and Profit Margin (Line) by Quarter

Problem 2 — Marketing Campaign Dashboard (5 Marks)

Given Data:

Channel	Spend (\$k)	Conversions
Email	2	150
Social	5	300
Search	8	550
Display	3	90

Task: Build a dashboard combining a bar chart of spend by channel and a scatter plot of spend vs. conversions. Annotate the most cost-efficient channel, and add a narrative caption recommending which channel to increase investment in.

Dashboard Design (Key Elements):

Chart Types	Bar chart (Spend by Channel) + Scatter/bubble plot (Spend vs. Conversions)
Shared Theme	Channel — same four categories and colour-coding used consistently across both panels
Aesthetic Mapping	Bar colour = Channel; scatter bubble colour = Channel, bubble size = Conversions
Annotation	Callout identifying Email as the most cost-efficient channel (150 conversions for only \$2k spend)
Narrative Takeaway	One-sentence recommendation on where to increase marketing investment

Design Explanation / Working:

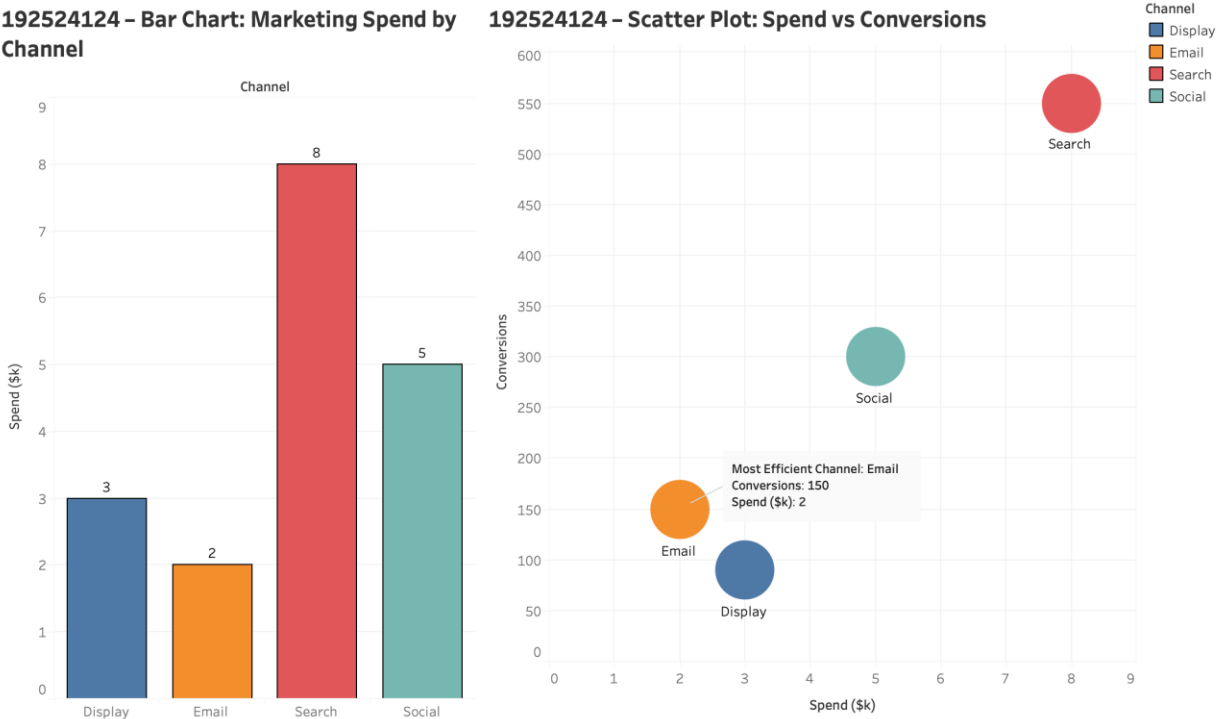
The left panel is a simple bar chart of spend by channel, giving a quick read of budget allocation, while the right panel is a scatter (bubble) plot with Spend on the X-axis and Conversions on the Y-axis, so a channel's cost and its outcome can be compared on two independent numeric scales at once rather than just one.

The same colour is used for each channel in both the bar chart and the scatter plot (e.g., Email is always orange, Search always red), which is what makes the two charts feel like one coordinated dashboard rather than two unrelated charts — a viewer can find a channel's bar, then immediately locate its matching-coloured bubble.

- Email: Spend = \$2k, Conversions = 150 → 75 conversions per \$1k spent — the highest efficiency ratio in the dataset.
- Social: Spend = \$5k, Conversions = 300 → 60 conversions per \$1k spent.
- Search: Spend = \$8k, Conversions = 550 → 68.75 conversions per \$1k spent, and the highest total conversions of any channel.
- Display: Spend = \$3k, Conversions = 90 → 30 conversions per \$1k spent — the least efficient channel.

Constructed Dashboard:

192524124 – Marketing Campaign Dashboard



Email delivers the highest conversions per \$1k spent, making it the most cost-efficient channel, while Search produces the highest total conversions and is a strong candidate for increased investment.

Figure 2: Marketing Campaign Dashboard — Spend by Channel (Bar) and Spend vs. Conversions (Scatter)

Problem 3 — Hospital Patient Flow Dashboard (5 Marks)

Given Data:

Day	Admissions	Avg Wait (min)
Mon	40	35
Tue	55	50
Wed	60	65
Thu	45	40
Fri	70	80

Task: Build a dashboard with a bar chart of admissions and a line chart of wait time on the same day axis. Add an annotation flagging the day needing urgent staffing attention, and write a one-sentence takeaway for hospital management.

Dashboard Design (Key Elements):

Chart Types	Bar chart (Daily Admissions) + Line chart (Average Wait Time)
Shared Axis	Day (Mon → Fri) — identical weekday axis on both panels
Aesthetic Mapping	Bar colour = Day; line-point colour intensity = Wait Time, reinforcing the upward trend
Annotation	Callout on Friday flagging it as needing urgent staffing (70 admissions, 80-minute wait)
Narrative Takeaway	One-sentence recommendation for hospital management on staffing

Design Explanation / Working:

Admissions are shown as a bar chart and average wait time as a line chart directly beside it, both sharing the same Monday-to-Friday axis so that any day's admission volume can be checked against its wait time without cross-referencing two separate tables.

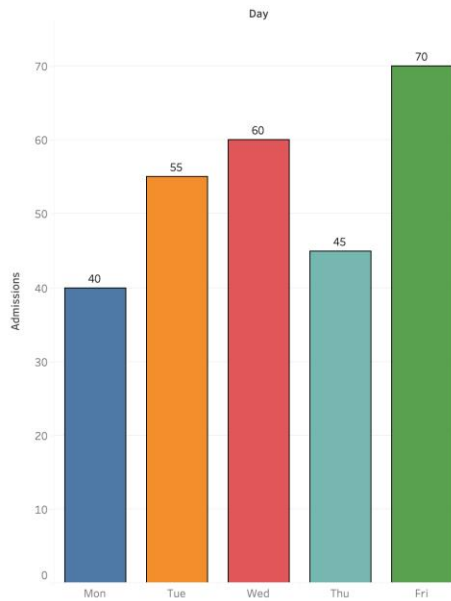
The line's markers use a light-to-dark colour gradient mapped to the wait-time value itself, so the visual weight of the line grows heavier toward the end of the week exactly as the bars grow taller — reinforcing, through colour alone, that admissions and wait time are rising together.

- Mon: Admissions = 40, Wait = 35 min — the lightest day.
- Tue: Admissions = 55, Wait = 50 min — both metrics rise.
- Wed: Admissions = 60, Wait = 65 min — wait time rises faster than admissions, a possible early bottleneck.
- Thu: Admissions = 45, Wait = 40 min — a brief dip in both metrics.
- Fri: Admissions = 70, Wait = 80 min — both metrics peak sharply, well above every other day.

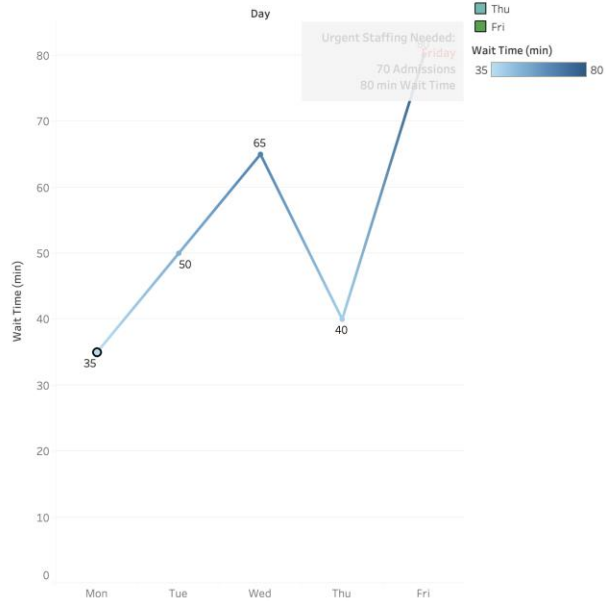
Constructed Dashboard:

192524124 – Hospital Patient Flow Dashboard

192524124 – Bar Chart: Daily Patient Admissions



192524124 – Line Chart: Average Wait Time



Friday experienced the highest patient admissions and the longest average wait time, indicating that additional staffing is needed to improve patient flow and reduce waiting times.

Admissions Bar Chart Wait Time Line Chart 192524124 – Hospital Pati...

Figure 3: Hospital Patient Flow Dashboard — Daily Admissions (Bar) and Average Wait Time (Line)

Problem 4 — School Performance Dashboard (5 Marks)

Given Data:

Grade	Avg Score (%)	Attendance (%)
Grade 6	72	91
Grade 7	68	85
Grade 8	75	93
Grade 9	60	78

Task: Build a dashboard combining a bar chart of average scores and a bar chart of attendance rate for the same grades. Annotate the grade needing intervention, and add a short narrative connecting attendance to performance.

Dashboard Design (Key Elements):

Chart Types	Bar chart (Average Test Score) + Bar chart (Attendance Rate)
Shared Axis	Grade (6 → 9) — identical grade axis and colour-coding on both panels
Aesthetic Mapping	Bar colour = Grade, kept consistent across both charts for direct visual matching
Annotation	Callout on Grade 9 identifying it as needing intervention (lowest score 60%, lowest attendance 78%)
Narrative Takeaway	One-sentence insight connecting attendance to academic performance

Design Explanation / Working:

Two bar charts are placed side by side — one for average test score, one for attendance rate — both using the same four grade categories (6 through 9) in the same order and the same colour for each grade, so a viewer can compare a grade's academic performance against its attendance simply by looking at the bar in the same horizontal position in each chart.

- Grade 6: Score = 72%, Attendance = 91% — highest attendance among the four grades, second-highest score.
- Grade 7: Score = 68%, Attendance = 85% — a mid-range result on both measures.
- Grade 8: Score = 75%, Attendance = 93% — the best result on both measures, the highest score and highest attendance.
- Grade 9: Score = 60%, Attendance = 78% — the lowest score and the lowest attendance, both by a clear margin.

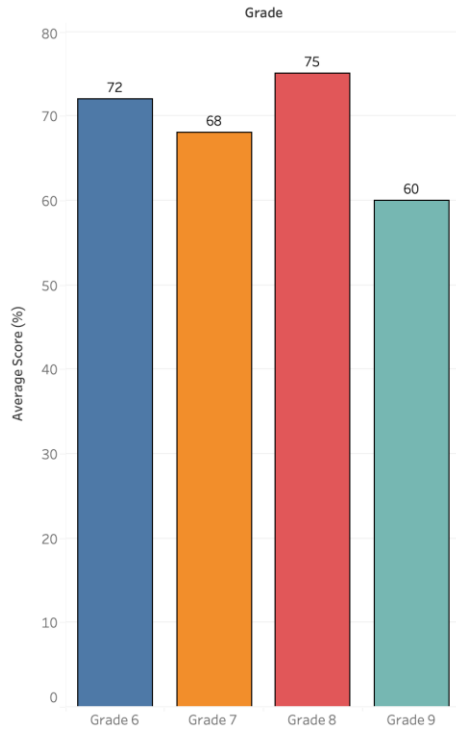
Because both charts consistently rank Grade 8 highest and Grade 9 lowest on both measures, the side-by-side design makes the score–attendance relationship visible without needing a third, more complex correlation chart.

The annotation is placed on Grade 9's attendance bar, explicitly labelling it 'Needs Intervention' and citing both of its lowest figures (Lowest Score: 60%, Lowest Attendance: 78%) so that the single grade requiring support is unambiguous at a glance.

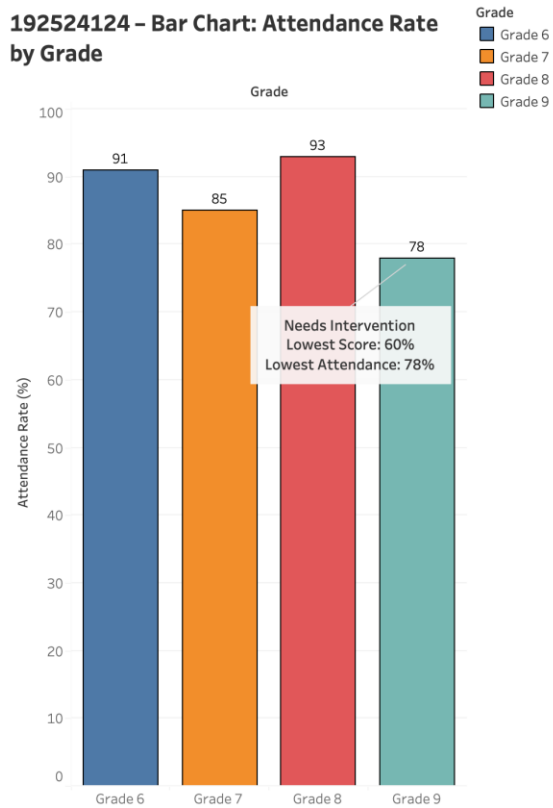
Constructed Dashboard:

192524124 – School Performance Dashboard

192524124 – Bar Chart: Average Test Scores by Grade



192524124 – Bar Chart: Attendance Rate by Grade



Grade 9 has both the lowest attendance rate and the lowest average test score, suggesting that improving attendance may help enhance academic performance.

Figure 4: School Performance Dashboard — Average Test Score and Attendance Rate by Grade

Problem 5 — E-commerce Product Dashboard (5 Marks)

Given Data:

Product	Units Sold	Return Rate (%)
A	500	5
B	350	12
C	600	3
D	200	20
E	450	8

Task: Build a dashboard with a bar chart of units sold and a bar chart of return rate for the same products, using a shared product axis and consistent colour-coding. Annotate the product that most urgently needs investigation, and write a one-sentence takeaway for the product team.

Dashboard Design (Key Elements):

Chart Types	Bar chart (Units Sold) + Bar chart (Return Rate)
Shared Axis	Product (A → E) — identical product axis and colour-coding on both panels
Aesthetic Mapping	Bar colour = Product, kept consistent across both charts for direct visual matching
Annotation	Callout on Product D identifying it as needing investigation (20% return rate, only 200 units sold)
Narrative Takeaway	One-sentence summary contrasting the best- and worst-performing products

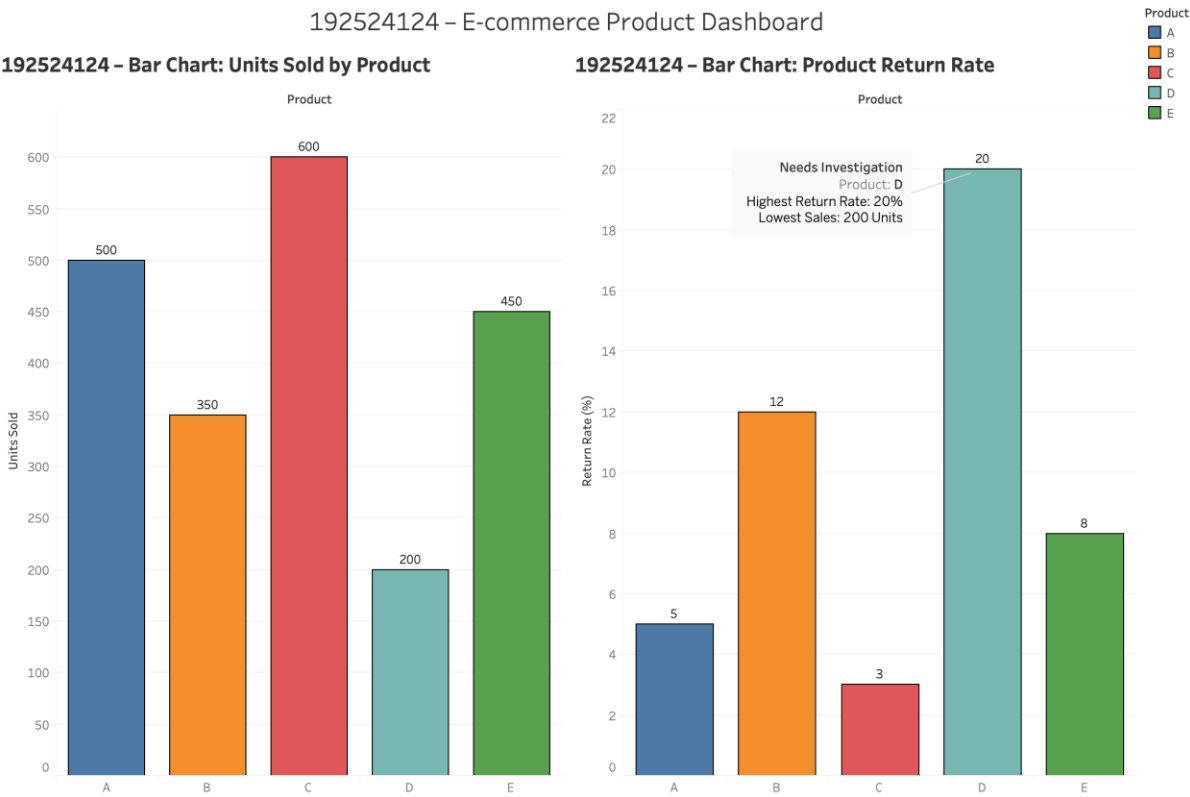
Design Explanation / Working:

Units sold and return rate are shown as two bar charts sharing the same five-product (A–E) axis and identical per-product colours, so the bar for any product in the sales chart lines up directly with that same product's bar in the return-rate chart.

- Product A: 500 units, 5% returns — solid sales, low returns.
- Product B: 350 units, 12% returns — moderate sales, elevated returns.
- Product C: 600 units, 3% returns — the best performer, highest sales and lowest return rate.
- Product D: 200 units, 20% returns — the worst performer, lowest sales and by far the highest return rate.
- Product E: 450 units, 8% returns — solid, mid-pack performance.

Reading the two charts together makes the underlying pattern immediately visible: Product C sits at one extreme (tall sales bar, short return bar) while Product D sits at the opposite extreme (short sales bar, tall return bar) — a relationship that would be far harder to spot from either chart alone.

Constructed Dashboard:



Product C has the highest sales with the lowest return rate, while Product D requires immediate attention due to its high return rate and low sales performance.

Figure 5: E-commerce Product Dashboard — Units Sold and Return Rate by Product